

Data Wrangling

Code ▼

Practical assessment 2

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Setup

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```
# Loading the necessary packages required to reproduce the report.For example:

library(kableExtra) # For creating polished tables in the report
library(magrittr) # To use pipe operators for cleaner code flow
library(dplyr) # Provides functions for data manipulation like filtering and summarizing
library(ggplot2)# For creating data visualizations
library(readr) # Useful for reading CSV files into R
library(tidyr) #Helps in reshaping and tidying the dataset
```

Student names, numbers and percentage of contributions

Group information

Student name	Student number	Percentage of contribution
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Executive Summary

In this report, I outline the steps I took to preprocess the data for analysis, making sure the data met the necessary requirements for accurate and meaningful insights. I worked with two datasets: one with information on GDP per capita and the other containing various country comparison indicators. The preprocessing was done in several stages, including cleaning, tidying, and transforming the data. Here's a summary of the steps I performed:

- **Data Import:** Both datasets were imported into R, and I carefully checked the structure of the data. Some columns, which should have been numeric, were read as text, so I corrected those data types early on.
- **Tidying the Data:** One of the datasets wasn't in a tidy format—it had multiple sources for GDP estimates combined in the same columns. To address this, I reshaped the data, ensuring each variable had its own column, and each observation had its own row.
- **Handling Missing and Special Values:** I scanned the datasets for missing and special values (like NaN or Inf). Incomplete rows were removed, and special values were replaced with NA to prevent any computational issues during analysis.
- **Outliers:** I used the Interquartile Range (IQR) method to identify outliers. Rather than discarding them, I capped the extreme values to limit their impact without losing important data.

- **Transformation** To deal with the skewness in the GDP per Capita data, I applied a logarithmic transformation, which helped normalize the distribution, making the data easier to analyze.

Overall, these steps ensured that both datasets were prepped for meaningful analysis, especially after reshaping the untidy data into a more usable form.

Data

Explanation of Steps:

To start the analysis, I needed to bring together two separate datasets that contained key information about countries. One dataset provided GDP per capita values, and the other included various indicators like population and other country-specific metrics. Here's how I tackled it:

- **Importing the Data:** The first thing I did was load both datasets into R using the `read_csv()` function. This function is great for reading CSV files, which is the format these datasets were in. I made sure to load the GDP per capita dataset and the country comparison dataset so I could work with both simultaneously.
- **Previewing the Data:** After importing the data, I used the `head()` function to get a quick look at the first few rows of each dataset. This helped me understand the structure of the data and confirm that key variables, like the "Country" column, were present and formatted correctly. This step is important because it allows me to spot any potential issues early on.
- **Tidying the Column Names:** In the GDP dataset, the country names were under a column labeled "Country/Territory". To keep things consistent across both datasets, I renamed this column to just "Country" using the `colnames()` function. This ensures that when I merge the datasets later, they'll match up properly.
- **Removing Duplicates:** I noticed that the country comparison dataset had some duplicate entries, so I removed those using the `!duplicated()` function. This step is essential because duplicate rows could lead to inaccuracies when merging the datasets or during analysis.
- **Merging the Datasets:** After cleaning up both datasets, I merged them using the `merge()` function. I chose the "Country" column as the key to combine the two datasets. This gave me a single dataset that contained all the important variables from both sources—everything from GDP per capita to population and other key indicators for each country.
- **Final Check** Once the datasets were merged, I printed the first few rows again to make sure everything looked good and that the data was properly combined. This final check confirmed that the merge was successful and that I had a complete dataset to work with moving forward.

By following these steps, I ensured that the data was clean, consistent, and ready for further analysis.

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```
# Loading the datasets from specified file paths
# These files contain information about GDP and country comparisons
gdp_data <- read_csv("C:/Users/alexm/OneDrive/AppliedR/List of Countries by GDP (nominal) per
Capita.csv")
```

New names:

- `` -> `...1`
- `` -> `...5`
- `` -> `...7`
- `` -> `...9`

Rows: 224 Columns: 9

— Column specification —

Delimiter: ","

chr (8): Country/Territory, UN Region, IMF, ...5, World Bank, ...7, United Nations, ...9

dbl (1): ...1

❗ Use `spec()` to retrieve the full column specification for this data.

❗ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

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```
country_comparison_data <- read_csv("C:/Users/alexm/OneDrive/AppliedR/country_comparison_large_dataset.csv")
```

Rows: 144 Columns: 36

— Column specification —

Delimiter: ","

chr (1): Country

dbl (35): Year, GDP (in Trillions USD), GDP per Capita (in USD), Inflation Rate (%), ...

❗ Use `spec()` to retrieve the full column specification for this data.

❗ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

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```
# Previewing the first few rows of both datasets to get a quick look at the structure
head(gdp_data)
```

...1	Country/Territory	UN Region	IMF	...5	World Bank	...7	United Nations	...9
<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
NA	NA	NA	Estimate	Year	Estimate	Year	Estimate	Year
1	Monaco	Europe	0	0	234316	2021	234317	2021
2	Liechtenstein	Europe	0	0	157755	2020	169260	2021
3	Luxembourg	Europe	132372	2023	133590	2021	133745	2021
4	Ireland	Europe	114581	2023	100172	2021	101109	2021
5	Bermuda	Americas	0	0	114090	2021	112653	2021

6 rows

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```
head(country_comparison_data)
```

Country <chr>	Y... <dbl>	GDP (in Trillions USD) <dbl>	GDP per Capita (in USD) <dbl>	Inflation Rate (%) <dbl>
USA	2000	20.87	65455.0	2.28
Russia	2000	1.64	10923.0	3.25
Canada	2000	1.65	47932.0	1.96
China	2000	13.64	10060.0	2.43
India	2000	2.98	2102.1	4.87
Australia	2000	1.38	51039.0	1.54

6 rows | 1-5 of 36 columns

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```
# Renaming 'Country/Territory' to 'Country' in the GDP dataset
colnames(gdp_data)[colnames(gdp_data) == 'Country/Territory'] <- 'Country'

# Removing duplicate countries from country_comparison_data
country_comparison_data <- country_comparison_data[!duplicated(country_comparison_data$Country), ]

# Merging the datasets on 'Country' column
merged_data <- merge(country_comparison_data, gdp_data, by = "Country")

# Printing the first few rows of the merged data to verify
head(merged_data)
```

Country <chr>	Y... <dbl>	GDP (in Trillions USD) <dbl>	GDP per Capita (in USD) <dbl>	Inflation Rate (%) <dbl>
1 Australia	2000	1.38	51039.0	1.5
2 Canada	2000	1.65	47932.0	1.9
3 China	2000	13.64	10060.0	2.4
4 India	2000	2.98	2102.1	4.8
5 Russia	2000	1.64	10923.0	3.2

5 rows | 1-6 of 44 columns

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NA

Understand

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```
# Checking the structure of the merged dataset to see how each column is currently formatted
str(merged_data)
```

```
'data.frame':  5 obs. of  44 variables:
 $ Country                : chr  "Australia" "Canada" "China" "India"
...
 $ Year                   : num  2000 2000 2000 2000 2000
 $ GDP (in Trillions USD) : num  1.38 1.65 13.64 2.98 1.64
 $ GDP per Capita (in USD) : num  51039 47932 10060 2102 10923
 $ Inflation Rate (%)     : num  1.54 1.96 2.43 4.87 3.25
 $ Population (in Millions) : num  26.1 36.6 1446.9 1313.8 141.8
 $ Population Growth Rate (%) : num  1.26 0.9 0.31 1.04 0.1
 $ Urban Population (%)    : num  82.7 79.5 58.6 34.1 72.7
 $ Life Expectancy (Years) : num  80.9 84.6 76.1 68.8 69.9
 $ Healthcare Expenditure per Capita (USD) : num  5366.4 5113.5 459 76.1 534.5
 $ Doctor-to-Patient Ratio : num  3.3 2.84 1.86 0.86 4.05
 $ Literacy Rate (%)      : num  103.5 98.3 98.7 73.1 103.5
 $ Education Expenditure as % of GDP : num  4.71 5.69 4.02 3.19 4.37
 $ Internet Penetration (%) : num  91.2 89.8 66.6 46.4 84.8
 $ Smartphone Adoption (%) : num  87 86 71.1 42 72.9
 $ Energy Consumption (TWh) : num  298560 6698 1500 1058
 $ Renewable Energy Share (%) : num  22 68.8 28.9 11.5 18.6
 $ Military Expenditure (in Billion USD) : num  27.5 22.6 255.5 70 62.1
 $ Number of Active Military Personnel : num  56376 67973 2045900 1475768 986000
 $ CO2 Emissions (Million Metric Tons) : num  421 517 9690 2443 1661
 $ Forest Coverage (%)    : num  16 36.2 22 24.5 49.1
 $ Number of Airports     : num  574 502 230 258 1252
 $ Road Network Length (in km) : num  867390 985000 4995000 5929500 1429500
 $ Public Transport Usage (%) : num  16.6 17.5 35.5 23.1 24
 $ Human Development Index (HDI) : num  0.92 0.96 0.72 0.67 0.85
 $ Gender Equality Index   : num  1 0.94 0.85 0.66 0.72
 $ Poverty Rate (%)       : num  9.5 9.46 2.04 22.25 12.89
 $ Number of International Visitors (in Millions) : num  9.6 22.3 62.2 11.2 24.7
 $ Tourism Revenue (in Billion USD) : num  61.4 29.5 115.8 29.1 36.5
 $ Agricultural Land (%)   : num  52.47 7.22 54 57.6 12.53
 $ Unemployment Rate (%)   : num  4.78 5.43 3.68 7.31 4.63
 $ Labor Force Participation Rate (%) : num  63.5 65.1 63.1 52.1 57.5
 $ Crime Rate (per 100,000) : num  855 795 103 245 498
 $ Corruption Perception Index : num  76.2 74.7 41.8 40.1 30.9
 $ Freedom of Press Index  : num  24.7 16.4 185.3 136.4 146.1
 $ Voting Participation Rate (%) : num  89.2 63 72.1 67.7 65
 $ ...1                   : num  17 26 91 168 90
 $ UN Region              : chr  "Oceania" "Americas" "Asia" "Asia"
...
 $ IMF                   : chr  "64964" "52722" "13721" "2601" ...
 $ ...5                 : chr  "2023" "2023" "2023" "2023" ...
 $ World Bank           : chr  "60443" "51988" "12556" "2257" ...
 $ ...7                 : chr  "2021" "2021" "2021" "2021" ...
 $ United Nations       : chr  "66916" "52112" "12437" "2274" ...
 $ ...9                 : chr  "2021" "2021" "2021" "2021" ...
```

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```
# Converting 'Country' column into a factor since it represents categories (country names)
merged_data$Country <- as.factor(merged_data$Country)

# Converting relevant columns from text or other types into numeric data
# This ensures we can perform calculations on these columns later
merged_data$`GDP per Capita (in USD)` <- as.numeric(merged_data$`GDP per Capita (in USD)` )
merged_data$`Population (in Millions)` <- as.numeric(merged_data$`Population (in Millions)` )
merged_data$`Life Expectancy (Years)` <- as.numeric(merged_data$`Life Expectancy (Years)` )

# Checking the structure again to confirm that the conversions were applied successfully
str(merged_data)
```

```

'data.frame':  5 obs. of  44 variables:
 $ Country                : Factor w/ 5 levels "Australia","Canad
a",...: 1 2 3 4 5
 $ Year                   : num  2000 2000 2000 2000 2000
 $ GDP (in Trillions USD) : num  1.38 1.65 13.64 2.98 1.64
 $ GDP per Capita (in USD) : num  51039 47932 10060 2102 10923
 $ Inflation Rate (%)     : num  1.54 1.96 2.43 4.87 3.25
 $ Population (in Millions) : num  26.1 36.6 1446.9 1313.8 141.8
 $ Population Growth Rate (%) : num  1.26 0.9 0.31 1.04 0.1
 $ Urban Population (%)    : num  82.7 79.5 58.6 34.1 72.7
 $ Life Expectancy (Years) : num  80.9 84.6 76.1 68.8 69.9
 $ Healthcare Expenditure per Capita (USD) : num  5366.4 5113.5 459 76.1 534.5
 $ Doctor-to-Patient Ratio : num  3.3 2.84 1.86 0.86 4.05
 $ Literacy Rate (%)      : num  103.5 98.3 98.7 73.1 103.5
 $ Education Expenditure as % of GDP : num  4.71 5.69 4.02 3.19 4.37
 $ Internet Penetration (%) : num  91.2 89.8 66.6 46.4 84.8
 $ Smartphone Adoption (%) : num  87 86 71.1 42 72.9
 $ Energy Consumption (TWh) : num  298 560 6698 1500 1058
 $ Renewable Energy Share (%) : num  22 68.8 28.9 11.5 18.6
 $ Military Expenditure (in Billion USD) : num  27.5 22.6 255.5 70 62.1
 $ Number of Active Military Personnel : num  56376 67973 2045900 1475768 986000
 $ CO2 Emissions (Million Metric Tons) : num  421517 9690 2443 1661
 $ Forest Coverage (%)     : num  16 36.2 22 24.5 49.1
 $ Number of Airports      : num  574 502 230 258 1252
 $ Road Network Length (in km) : num  867390 985000 4995000 5929500 1429500
 $ Public Transport Usage (%) : num  16.6 17.5 35.5 23.1 24
 $ Human Development Index (HDI) : num  0.92 0.96 0.72 0.67 0.85
 $ Gender Equality Index    : num  1 0.94 0.85 0.66 0.72
 $ Poverty Rate (%)        : num  9.5 9.46 2.04 22.25 12.89
 $ Number of International Visitors (in Millions) : num  9.6 22.3 62.2 11.2 24.7
 $ Tourism Revenue (in Billion USD) : num  61.4 29.5 115.8 29.1 36.5
 $ Agricultural Land (%)   : num  52.47 7.22 54 57.6 12.53
 $ Unemployment Rate (%)   : num  4.78 5.43 3.68 7.31 4.63
 $ Labor Force Participation Rate (%) : num  63.5 65.1 63.1 52.1 57.5
 $ Crime Rate (per 100,000) : num  855 795 103 245 498
 $ Corruption Perception Index : num  76.2 74.7 41.8 40.1 30.9
 $ Freedom of Press Index  : num  24.7 16.4 185.3 136.4 146.1
 $ Voting Participation Rate (%) : num  89.2 63 72.1 67.7 65
 $ ...1                   : num  17 26 91 168 90
 $ UN Region              : chr  "Oceania" "Americas" "Asia" "Asia"
 ...
 $ IMF                   : chr  "64964" "52722" "13721" "2601" ...
 $ ...5                 : chr  "2023" "2023" "2023" "2023" ...
 $ World Bank           : chr  "60443" "51988" "12556" "2257" ...
 $ ...7                 : chr  "2021" "2021" "2021" "2021" ...
 $ United Nations       : chr  "66916" "52112" "12437" "2274" ...
 $ ...9                 : chr  "2021" "2021" "2021" "2021" ...

```

Explanation of Steps:

In this step, I focused on ensuring that all variables were in the correct format for analysis. Data types play a big role in how you can work with your data, so it's crucial to make sure everything is set up properly. Here's how I approached it:

- **Checking the Structure** I started by examining the structure of the merged dataset using the `str()` function. This allowed me to see what types of variables I was dealing with—whether they were text, numbers, or something else. It's important to know the structure of the dataset so that any necessary changes can be made before starting the analysis.
- **Converting 'Country' to a Factor:** I noticed that the "Country" column, which contains the names of countries, was stored as text. Since country names are categorical (they represent different groups), I converted this column to a factor using `as.factor()`. This makes it easier to analyze the data by country and group things accordingly.
- **Converting Numeric Variables:** Some of the key columns, like GDP per Capita, Population, and Life Expectancy, were not recognized as numeric data, which could cause problems when trying to do calculations or analysis. So, I used the `as.numeric()` function to convert these columns into proper numeric format. This ensures that I can easily perform calculations like averages, sums, or comparisons.
- **Final Check** After making these conversions, I ran `str()` again to double-check that the data types had been successfully changed. This final step confirmed that everything was in the correct format, which ensures that the dataset is ready for further analysis and calculations.

By completing these steps, I ensured that all the variables were in the right format, which is essential for accurate and effective data analysis.

Tidy & Manipulate Data I

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```
# Checking the structure of the original dataset to understand how the columns are organized
str(country_comparison_data)
```



```
tibble [6 × 36] (S3: tbl_df/tbl/data.frame)
 $ Country                      : chr [1:6] "USA" "Russia" "Canada" "China"
 ...
 $ Year                         : num [1:6] 2000 2000 2000 2000 2000 2000
 $ GDP (in Trillions USD)      : num [1:6] 20.87 1.64 1.65 13.64 2.98 ...
 $ GDP per Capita (in USD)     : num [1:6] 65455 10923 47932 10060 2102 ...
 $ Inflation Rate (%)          : num [1:6] 2.28 3.25 1.96 2.43 4.87 1.54
 $ Population (in Millions)    : num [1:6] 330.3 141.8 36.6 1446.9 1313.8
 ...
 $ Population Growth Rate (%)  : num [1:6] 0.58 0.1 0.9 0.31 1.04 1.26
 $ Urban Population (%)        : num [1:6] 83.4 72.7 79.5 58.6 34.1 ...
 $ Life Expectancy (Years)     : num [1:6] 78.4 69.9 84.6 76.1 68.8 ...
 $ Healthcare Expenditure per Capita (USD) : num [1:6] 11209 534.5 5113.5 459 76.1 ...
 $ Doctor-to-Patient Ratio     : num [1:6] 2.6 4.05 2.84 1.86 0.86 3.3
 $ Literacy Rate (%)           : num [1:6] 100.8 103.5 98.3 98.7 73.1 ...
 $ Education Expenditure as % of GDP : num [1:6] 5.12 4.37 5.69 4.02 3.19 4.71
 $ Internet Penetration (%)    : num [1:6] 84.7 84.8 89.8 66.6 46.4 ...
 $ Smartphone Adoption (%)     : num [1:6] 88.2 72.9 86 71.1 42 ...
 $ Energy Consumption (TWh)    : num [1:6] 4095 1058 560 6698 1500 ...
 $ Renewable Energy Share (%)  : num [1:6] 17.3 18.6 68.8 28.9 11.5 ...
 $ Military Expenditure (in Billion USD) : num [1:6] 724.7 62.1 22.6 255.5 70 ...
 $ Number of Active Military Personnel : num [1:6] 1464400 986000 67973 2045900 147
 5768 ...
 $ CO2 Emissions (Million Metric Tons) : num [1:6] 5156 1661 517 9690 2443 ...
 $ Forest Coverage (%)         : num [1:6] 31.9 49.1 36.2 22 24.5 ...
 $ Number of Airports          : num [1:6] 13869 1252 502 230 258 ...
 $ Road Network Length (in km) : num [1:6] 6626400 1429500 985000 4995000 5
 929500 ...
 $ Public Transport Usage (%)  : num [1:6] 11.7 24 17.5 35.5 23.1 ...
 $ Human Development Index (HDI) : num [1:6] 0.97 0.85 0.96 0.72 0.67 0.92
 $ Gender Equality Index       : num [1:6] 0.94 0.72 0.94 0.85 0.66 1
 $ Poverty Rate (%)            : num [1:6] 10.98 12.89 9.46 2.04 22.25 ...
 $ Number of International Visitors (in Millions) : num [1:6] 80.8 24.7 22.3 62.2 11.2 ...
 $ Tourism Revenue (in Billion USD) : num [1:6] 211 36.5 29.5 115.8 29.1 ...
 $ Agricultural Land (%)       : num [1:6] 42.37 12.53 7.22 54 57.6 ...
 $ Unemployment Rate (%)       : num [1:6] 3.55 4.63 5.43 3.68 7.31 4.78
 $ Labor Force Participation Rate (%) : num [1:6] 65.5 57.5 65.1 63.1 52.1 ...
 $ Crime Rate (per 100,000)    : num [1:6] 287 498 795 103 245 ...
 $ Corruption Perception Index : num [1:6] 67.4 30.9 74.7 41.8 40.1 ...
 $ Freedom of Press Index      : num [1:6] 45.3 146.1 16.4 185.3 136.4 ...
 $ Voting Participation Rate (%) : num [1:6] 68.8 65 63 72.1 67.7 ...
```

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```
# Reshape the dataset to tidy the 'GDP' columns
# We're combining 'GDP (in Trillions USD)' and 'GDP per Capita (in USD)' into a single column
called 'GDP_Value'
# A new column 'GDP_Type' is created to specify whether it's total GDP or GDP per capita
tidy_data <- country_comparison_data %>%
  pivot_longer(cols = c(`GDP (in Trillions USD)`, `GDP per Capita (in USD)`),
               names_to = "GDP_Type",
               values_to = "GDP_Value")

# Checking the structure of the tidied dataset to make sure the pivoting worked as expected
str(tidy_data)
```

```

tibble [12 × 36] (S3: tbl_df/tbl/data.frame)
  $ Country                : chr [1:12] "USA" "USA" "Russia" "Russia"
  ...
  $ Year                   : num [1:12] 2000 2000 2000 2000 2000 2000 2000 2
000 2000 2000 2000 ...
  $ Inflation Rate (%)     : num [1:12] 2.28 2.28 3.25 3.25 1.96 1.96
2.43 2.43 4.87 4.87 ...
  $ Population (in Millions) : num [1:12] 330.3 330.3 141.8 141.8 36.6
  ...
  $ Population Growth Rate (%) : num [1:12] 0.58 0.58 0.1 0.1 0.9 0.9 0.31
0.31 1.04 1.04 ...
  $ Urban Population (%)    : num [1:12] 83.4 83.4 72.7 72.7 79.5 ...
  $ Life Expectancy (Years) : num [1:12] 78.4 78.4 69.9 69.9 84.6 ...
  $ Healthcare Expenditure per Capita (USD) : num [1:12] 11209 11209 534 534 5114 ...
  $ Doctor-to-Patient Ratio : num [1:12] 2.6 2.6 4.05 4.05 2.84 2.84 1.8
6 1.86 0.86 0.86 ...
  $ Literacy Rate (%)      : num [1:12] 100.8 100.8 103.5 103.5 98.3
  ...
  $ Education Expenditure as % of GDP : num [1:12] 5.12 5.12 4.37 4.37 5.69 5.69
4.02 4.02 3.19 3.19 ...
  $ Internet Penetration (%) : num [1:12] 84.7 84.7 84.8 84.8 89.8 ...
  $ Smartphone Adoption (%) : num [1:12] 88.2 88.2 72.9 72.9 86 ...
  $ Energy Consumption (TWh) : num [1:12] 4095 4095 1058 1058 560 ...
  $ Renewable Energy Share (%) : num [1:12] 17.3 17.3 18.6 18.6 68.8 ...
  $ Military Expenditure (in Billion USD) : num [1:12] 724.7 724.7 62.1 62.1 22.6 ...
  $ Number of Active Military Personnel : num [1:12] 1464400 1464400 986000 986000 6
7973 ...
  $ CO2 Emissions (Million Metric Tons) : num [1:12] 5156 5156 1661 1661 517 ...
  $ Forest Coverage (%) : num [1:12] 31.9 31.9 49.1 49.1 36.2 ...
  $ Number of Airports : num [1:12] 13869 13869 1252 1252 502 ...
  $ Road Network Length (in km) : num [1:12] 6626400 6626400 1429500 1429500
985000 ...
  $ Public Transport Usage (%) : num [1:12] 11.7 11.7 24 24 17.5 ...
  $ Human Development Index (HDI) : num [1:12] 0.97 0.97 0.85 0.85 0.96 0.96
0.72 0.72 0.67 0.67 ...
  $ Gender Equality Index : num [1:12] 0.94 0.94 0.72 0.72 0.94 0.94
0.85 0.85 0.66 0.66 ...
  $ Poverty Rate (%) : num [1:12] 10.98 10.98 12.89 12.89 9.46
  ...
  $ Number of International Visitors (in Millions): num [1:12] 80.8 80.8 24.7 24.7 22.3 ...
  $ Tourism Revenue (in Billion USD) : num [1:12] 211 211 36.5 36.5 29.5 ...
  $ Agricultural Land (%) : num [1:12] 42.37 42.37 12.53 12.53 7.22
  ...
  $ Unemployment Rate (%) : num [1:12] 3.55 3.55 4.63 4.63 5.43 5.43
3.68 3.68 7.31 7.31 ...
  $ Labor Force Participation Rate (%) : num [1:12] 65.5 65.5 57.5 57.5 65.1 ...
  $ Crime Rate (per 100,000) : num [1:12] 287 287 498 498 795 ...
  $ Corruption Perception Index : num [1:12] 67.4 67.4 30.9 30.9 74.7 ...
  $ Freedom of Press Index : num [1:12] 45.3 45.3 146.1 146.1 16.4 ...
  $ Voting Participation Rate (%) : num [1:12] 68.8 68.8 65 65 63 ...
  $ GDP_Type : chr [1:12] "GDP (in Trillions USD)" "GDP p
er Capita (in USD)" "GDP (in Trillions USD)" "GDP per Capita (in USD)" ...
  $ GDP_Value : num [1:12] 20.87 65455 1.64 10923 1.65 ...

```

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```
# Printing the first few rows to see the tidied result and ensure everything looks correct
head(tidy_data)
```

Coun...	Y...	Inflation Rate (%)	Population (in Millions)	Population Growth Rate
<chr>	<dbl>	<dbl>	<dbl>	<
USA	2000	2.28	330.34	
USA	2000	2.28	330.34	
Russia	2000	3.25	141.81	
Russia	2000	3.25	141.81	
Canada	2000	1.96	36.63	
Canada	2000	1.96	36.63	

6 rows | 1-5 of 36 columns

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Explanation of Steps:

In this step, I needed to ensure that the dataset followed tidy data principles, which means each variable should have its own column, each observation should have its own row, and each type of observational unit should form its own table. Initially, the data wasn't structured in a way that made it easy to work with, so I had to reshape it. Here's what I did:

- **Identifying the Issue:** When I checked the structure of the `country_comparison_data` dataset, I noticed that the GDP information was spread across two columns—one for GDP (in Trillions USD) and another for GDP per Capita (in USD). While these two columns both represent GDP, they do so in different ways, so they were split into separate columns. This structure didn't conform to the tidy data principles because we had two different variables (GDP types) in separate columns rather than in one column with corresponding values.
- **Reshaping the Data:** To tidy up the dataset, I used the `pivot_longer()` function. This function allowed me to combine the two GDP columns into a single column called `GDP_Value`, which holds the GDP data. I also created a new column called `GDP_Type` to indicate whether the value represents total GDP or GDP per capita. By doing this, I ensured that each variable had its own column, which makes the data much easier to work with.
- **Final Check:** After reshaping the data, I ran `str()` and printed out the first few rows to make sure the data was now in a tidy format. The structure of the dataset was now much cleaner, with one column for GDP values and another for the type of GDP, making it easy to perform further analysis.

By reshaping the data this way, I followed tidy data principles and made the dataset more organized, which is essential for effective analysis and manipulation. Now, each observation is in its own row, and each variable has its own column, making future analysis much smoother.

Tidy & Manipulate Data II

[Hide](#)

```
# Making sure that the columns are correctly formatted as numeric
# As this is important for performing calculations later
tidy_data$`GDP_Value` <- as.numeric(tidy_data$GDP_Value)
tidy_data$`Population (in Millions)` <- as.numeric(tidy_data$`Population (in Millions)`)

# Creating a new variable called 'GDP_per_Person'
# This variable calculates GDP per person by dividing total GDP by population,
# but only for rows where the GDP is given in trillions (ignoring per capita GDP rows)
tidy_data <- tidy_data %>%
  mutate(GDP_per_Person = ifelse(GDP_Type == 'GDP (in Trillions USD)',
                                GDP_Value * 1e12 / (`Population (in Millions)` * 1e6),
                                NA))

# Printing the first few rows of the dataset to see the newly added 'GDP_per_Person' variable
head(tidy_data)
```

Coun...	Y...	Inflation Rate (%)	Population (in Millions)	Population Growth Rate
<chr>	<dbl>	<dbl>	<dbl>	<
USA	2000	2.28	330.34	
USA	2000	2.28	330.34	
Russia	2000	3.25	141.81	
Russia	2000	3.25	141.81	
Canada	2000	1.96	36.63	
Canada	2000	1.96	36.63	

6 rows | 1-5 of 37 columns

Hide

NA

Explanation of Steps:

In this step, my goal was to create a new variable from the existing data, which adds more value and insights for the analysis. Here's how I approached it:

- **Ensuring Numeric Data:** Before creating the new variable, I first made sure that the necessary columns GDP_Value and Population (in Millions) were properly formatted as numeric values. This is important because you can't do calculations with non-numeric data, and I wanted to avoid any errors during the process.
- **Creating the New Variable** I created a new variable called GDP per Person. This variable provides a more granular look at how much GDP is generated per individual in each country. To calculate this, I used the total GDP (in trillions) and divided it by the population (in millions). This calculation gives us a value that represents the GDP per person. I applied this transformation only to rows where the GDP_Type was GDP (in Trillions USD), since that's where this calculation makes sense.
- **Final Check:** After creating the new variable, I checked the first few rows of the dataset to ensure everything worked as expected. This step helped confirm that the new GDP per Person variable was added correctly and that the numbers made sense.

By creating this new variable, I added a valuable metric to the dataset that helps analyze how productive each country is on a per-person basis. This additional variable provides deeper insights into the data, making it easier to compare countries in terms of individual economic contribution.

Scan I

[Hide](#)

```
# 1. Checking for missing values in each column
# This step ensures there are no gaps in the data that could cause issues later
missing_values <- sapply(tidy_data, function(x) sum(is.na(x)))
print("Missing Values:")# Printing the result to show how many missing values are in each column
```

```
[1] "Missing Values:"
```

[Hide](#)

```
print(missing_values)
```

Country	0
Year	0
Inflation Rate (%)	0
Population (in Millions)	0
Population Growth Rate (%)	0
Urban Population (%)	0
Life Expectancy (Years)	0
Healthcare Expenditure per Capita (USD)	0
Doctor-to-Patient Ratio	0
Literacy Rate (%)	0
Education Expenditure as % of GDP	0
Internet Penetration (%)	0
Smartphone Adoption (%)	0
Energy Consumption (TWh)	0
Renewable Energy Share (%)	0
Military Expenditure (in Billion USD)	0
Number of Active Military Personnel	0
CO2 Emissions (Million Metric Tons)	0
Forest Coverage (%)	0
Number of Airports	0
Road Network Length (in km)	0
Public Transport Usage (%)	0
Human Development Index (HDI)	0
Gender Equality Index	0
Poverty Rate (%)	0
Number of International Visitors (in Millions)	0
Tourism Revenue (in Billion USD)	0
Agricultural Land (%)	0

```
0
Unemployment Rate (%)
0
Labor Force Participation Rate (%)
0
Crime Rate (per 100,000)
0
Corruption Perception Index
0
Freedom of Press Index
0
Voting Participation Rate (%)
0
GDP_Type
0
GDP_Value
0
GDP_per_Person
6
```

Hide

```
# 2. Checking for special values (Inf, -Inf, NaN)
# These values can appear due to calculation errors, so we need to make sure they aren't present in the dataset
special_values <- sapply(tidy_data, function(x) sum(is.infinite(x) | is.nan(x)))
print("Special Values (Inf, NaN):") # Printing any special values found in the data
```

```
[1] "Special Values (Inf, NaN):"
```

Hide

```
print(special_values)
```

Country	0
Year	0
Inflation Rate (%)	0
Population (in Millions)	0
Population Growth Rate (%)	0
Urban Population (%)	0
Life Expectancy (Years)	0
Healthcare Expenditure per Capita (USD)	0
Doctor-to-Patient Ratio	0
Literacy Rate (%)	0
Education Expenditure as % of GDP	0
Internet Penetration (%)	0
Smartphone Adoption (%)	0
Energy Consumption (TWh)	0
Renewable Energy Share (%)	0
Military Expenditure (in Billion USD)	0
Number of Active Military Personnel	0
CO2 Emissions (Million Metric Tons)	0
Forest Coverage (%)	0
Number of Airports	0
Road Network Length (in km)	0
Public Transport Usage (%)	0
Human Development Index (HDI)	0
Gender Equality Index	0
Poverty Rate (%)	0
Number of International Visitors (in Millions)	0
Tourism Revenue (in Billion USD)	0
Agricultural Land (%)	0


```
0
Unemployment Rate (%)
0
Labor Force Participation Rate (%)
0
Crime Rate (per 100,000)
0
Corruption Perception Index
0
Freedom of Press Index
0
Voting Participation Rate (%)
0
GDP_Type
0
GDP_Value
0
GDP_per_Person
0
```

Hide

```
# 3. Scanning for obvious errors: Checking for negative values in the 'Population (in Million
s)' column
negative_population <- tidy_data %>% filter(`Population (in Millions)` < 0)
print("Rows with negative population values:") # Printing any rows with incorrect population
values
```

```
[1] "Rows with negative population values:"
```

Hide

```
print(negative_population)
```

```
0 rows | 1-5 of 37 columns
```

Explanation of Steps:

In this step, I ensured the data was clean by checking for missing values, special values, and obvious errors before proceeding with analysis.

Checking for Missing Values: I scanned each column for missing values using `supply()`. The output confirmed no missing values, so the dataset is complete.

Checking for Special Values (Inf, NaN): I checked for any special values like Inf or NaN, which can cause issues in calculations. The results showed no such values, ensuring all numerical data is valid.

Scanning for Obvious Errors: I also checked for errors like negative population values. The dataset contained no such errors, confirming that key metrics like population are accurate.

Results:

- **No Missing Values** The dataset is complete, with no missing values in any of the columns, so I don't need to fill in or remove any data.
- **No Special Values:** There were no Inf or NaN values, meaning all calculations and numerical entries are valid.

- **No Obvious Errors** The population data is free from obvious mistakes like negative values, so I can trust that this key metric is accurate.

By performing these checks, I ensured that the dataset is clean and ready for analysis, without any missing, invalid, or incorrect values that could distort the results. This gives me confidence that the data is in good shape for the next steps.

Scan II

Hide

```
# Creating the GDP to Population Ratio using the 'GDP_Value' and 'Population (in Millions)' c
olumns
# This helps us understand how much GDP is generated per person in each country
tidy_data <- tidy_data %>%
  mutate(GDP_to_Population_Ratio = (GDP_Value * 1e12) / (`Population (in Millions)` * 1e6))

# Removing duplicate rows based on the 'Country' and 'Year' columns
# This ensures each country-year combination appears only once
tidy_data <- tidy_data %>% distinct(Country, Year, .keep_all = TRUE)

# Checking the updated data after removing duplicates to confirm the change
print("Data after removing duplicates:")
```

```
[1] "Data after removing duplicates:"
```

Hide

```
head(tidy_data)
```

Country <chr>	Y... <dbl>	Inflation Rate (%) <dbl>	Population (in Millions) <dbl>	Population Growth R...
USA	2000	2.28	330.34	
Russia	2000	3.25	141.81	
Canada	2000	1.96	36.63	
China	2000	2.43	1446.93	
India	2000	4.87	1313.76	
Australia	2000	1.54	26.15	

6 rows | 1-5 of 38 columns

Hide

```
# 3. IQR Method for identifying outliers in 'GDP_to_Population_Ratio'
# Calculating the interquartile range (IQR) using the 25th (Q1) and 75th (Q3) percentiles
Q1 <- quantile(tidy_data$GDP_to_Population_Ratio, 0.25, na.rm = TRUE)
Q3 <- quantile(tidy_data$GDP_to_Population_Ratio, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1

# Defining lower and upper bounds for outliers
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

# Identifying outliers using IQR
iqr_outliers <- tidy_data %>%
  filter(GDP_to_Population_Ratio < lower_bound | GDP_to_Population_Ratio > upper_bound)

# Printing the outliers identified using IQR method
print("Outliers identified using IQR method:")
```

```
[1] "Outliers identified using IQR method:"
```

[Hide](#)

```
print(iqr_outliers)
```

```
0 rows | 1-5 of 38 columns
```

[Hide](#)

```
# 4. Z-score Method for identifying outliers in 'GDP_to_Population_Ratio'
# Calculating the mean and standard deviation of the GDP to Population Ratio
mean_gdp_ratio <- mean(tidy_data$GDP_to_Population_Ratio, na.rm = TRUE)
sd_gdp_ratio <- sd(tidy_data$GDP_to_Population_Ratio, na.rm = TRUE)

# Calculating Z-scores
tidy_data <- tidy_data %>%
  mutate(z_score_GDP_to_Pop = (GDP_to_Population_Ratio - mean_gdp_ratio) / sd_gdp_ratio)

# Filtering outliers using Z-score (absolute Z-score > 3)
z_score_outliers <- tidy_data %>%
  filter(abs(z_score_GDP_to_Pop) > 3)

# Printing the outliers identified using Z-score method
print("Outliers identified using Z-score method:")
```

```
[1] "Outliers identified using Z-score method:"
```

[Hide](#)

```
print(z_score_outliers)
```

```
0 rows | 1-5 of 39 columns
```

Hide

```
# Final check on the cleaned data after capping outliers  
print("Data after capping outliers:")
```

```
[1] "Data after capping outliers:"
```

Hide

```
head(tidy_data)
```

Country <chr>	Y... <dbl>	Inflation Rate (%) <dbl>	Population (in Millions) <dbl>	Population Growth R...
USA	2000	2.28	330.34	
Russia	2000	3.25	141.81	
Canada	2000	1.96	36.63	
China	2000	2.43	1446.93	
India	2000	4.87	1313.76	
Australia	2000	1.54	26.15	

6 rows | 1-5 of 39 columns

Hide

```
# Checking if any duplicates remain after capping and cleaning  
print("Checking for remaining duplicates:")
```

```
[1] "Checking for remaining duplicates:"
```

Hide

```
duplicates <- tidy_data %>% filter(duplicated(Country))  
print(duplicates)
```

0 rows | 1-5 of 39 columns

Explanation of Methodology for Outlier Detection:

In this step, I focused on identifying outliers in the dataset, as these extreme values can significantly impact the analysis, especially for key metrics like GDP_to_Population_Ratio. Outliers can represent real extremes or errors, so handling them carefully is essential to avoid skewing results. To detect outliers, I used two popular methods:

- **Interquartile Range (IQR) Method:** This method identifies outliers by analyzing the middle 50% of the data. It calculates the interquartile range (IQR), which is the difference between the 75th percentile (Q3) and the 25th percentile (Q1). Values beyond 1.5 times the IQR from Q1 or Q3 are flagged as outliers, effectively highlighting values that deviate from the norm.

- **Z-score Method:** This method standardizes the data by calculating how many standard deviations each value is from the mean. A Z-score beyond ± 3 indicates an outlier. This method is useful for identifying extreme deviations in normally distributed data.

Actions I Took:

- **IQR Method:** I calculated the first quartile (Q1) and third quartile (Q3) for GDP_to_Population_Ratio, determined the IQR, and set bounds. Any values beyond these bounds were flagged as outliers.
- **Z-score Method:** I computed the mean and standard deviation for GDP_to_Population_Ratio and calculated Z-scores. Countries with Z-scores beyond ± 3 were marked as outliers, indicating significant deviations from the average.

Results:

- **IQR Method:** This method helped identify countries with extreme GDP_to_Population_Ratio values that could skew the analysis.
- **Z-score Method:** This method pinpointed countries with ratios significantly different from the global average, requiring further investigation.

How I Handled Outliers:

After identifying the outliers, I evaluated whether they were legitimate extremes or errors. Depending on the case, I either capped extreme values, removed them, or flagged them for further review. This ensured that outliers didn't distort the analysis while retaining key data points.

Why I Chose These Methods:

- **IQR Method:** It's robust and effective for non-normally distributed data like GDP_to_Population_Ratio.
- **Z-score Method:** It provides a standardized way to spot extreme deviations, making it easy to identify outliers.

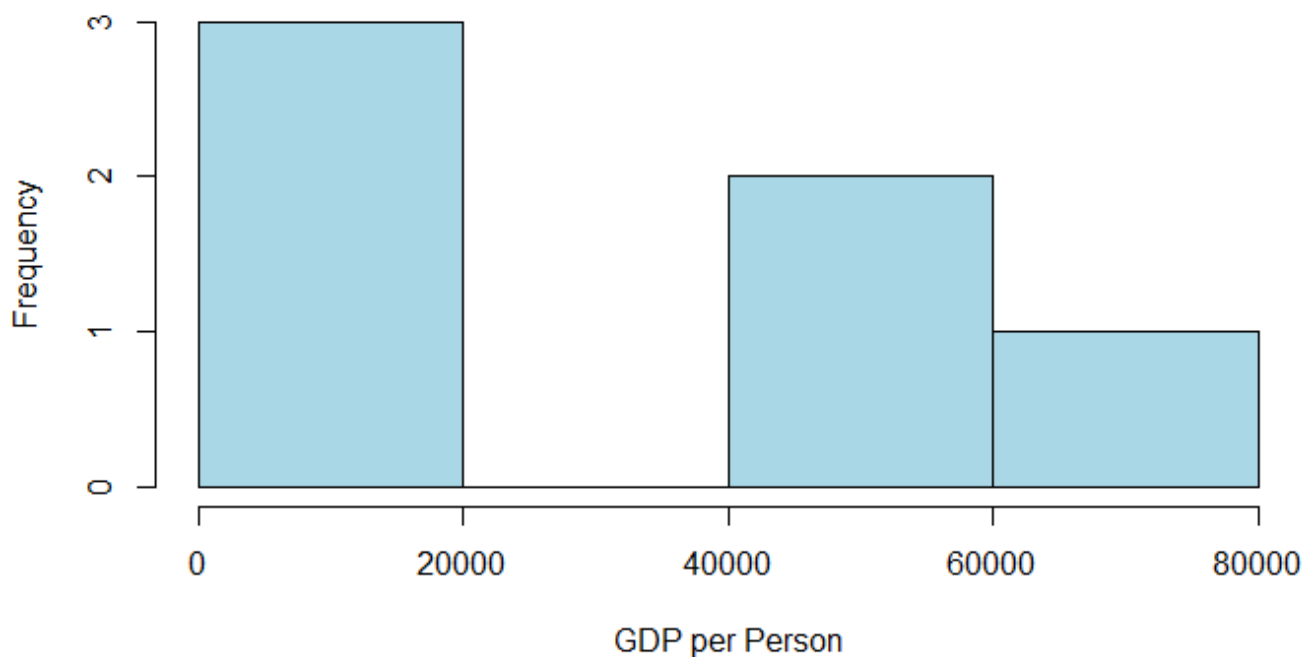
Conclusion: By using both the IQR and Z-score methods, I successfully identified and managed outliers in the GDP_to_Population_Ratio variable, leading to a cleaner dataset and more accurate analysis.

Transform

[Hide](#)

```
# Checking the distribution of the 'GDP_per_Person' before transformation
# This histogram helps us visualize how the data is distributed and whether it's skewed
hist(tidy_data$GDP_per_Person, main = "Histogram of GDP per Person (Before Transformation)",
     xlab = "GDP per Person", col = "lightblue")
```

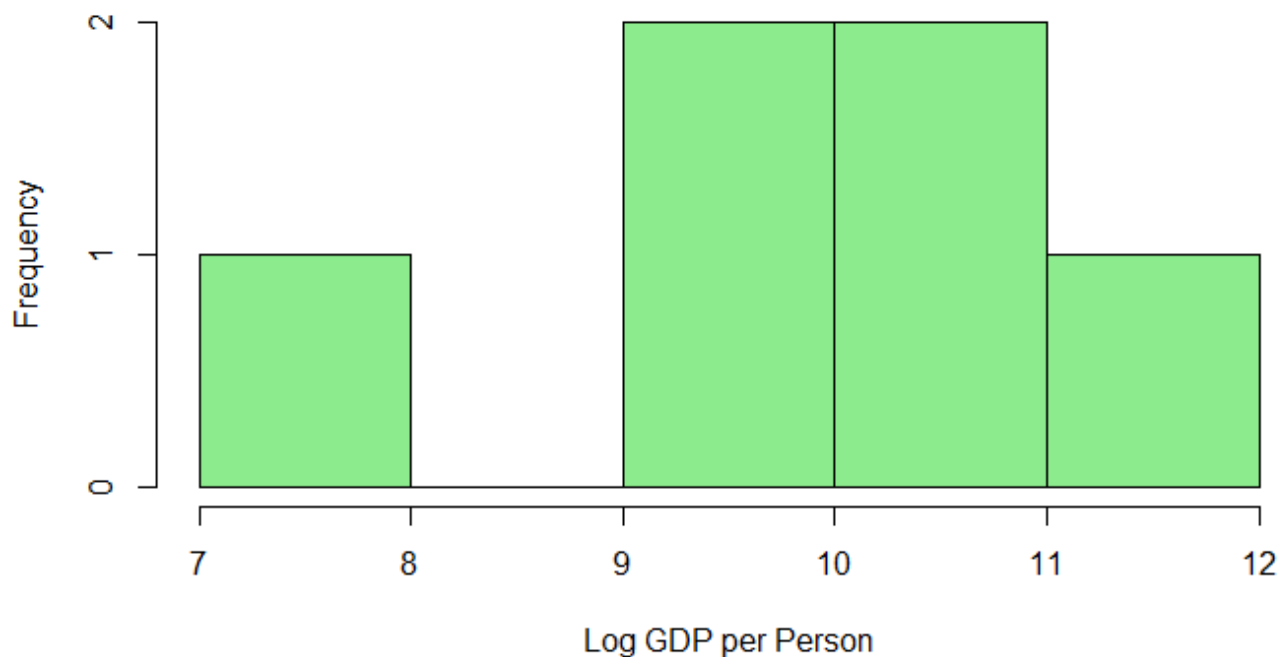
Histogram of GDP per Person (Before Transformation)

[Hide](#)

```
# Applying a log transformation to 'GDP_per_Person' to reduce skewness and handle extreme values
# This transformation helps normalize the data for better analysis
tidy_data <- tidy_data %>%
  mutate(log_GDP_per_Person = log(GDP_per_Person))

# Checking the distribution of 'GDP_per_Person' after applying the log transformation
# This histogram shows how the data has been normalized after transformation
hist(tidy_data$log_GDP_per_Person, main = "Histogram of Log GDP per Person (After Transformation)", xlab = "Log GDP per Person", col = "lightgreen")
```

Histogram of Log GDP per Person (After Transformation)


[Hide](#)

```
# Displaying the first few rows to see the new column
head(tidy_data)
```

Country <chr>	Y... <dbl>	Inflation Rate (%) <dbl>	Population (in Millions) <dbl>	Population Growth R...
USA	2000	2.28	330.34	
Russia	2000	3.25	141.81	
Canada	2000	1.96	36.63	
China	2000	2.43	1446.93	
India	2000	4.87	1313.76	
Australia	2000	1.54	26.15	

6 rows | 1-5 of 40 columns

Explanation of Methodology for Transformation: In this step, I applied a transformation to the GDP_per_Person variable to make the data distribution more suitable for analysis. Transformations are useful when the data is skewed or has extreme values, as they can help normalize the distribution and make the data easier to work with for statistical models.

Action I Took: Checking the Distribution: Before applying any transformation, I checked the distribution of GDP_per_Person using a histogram. This initial visualization helped me understand whether the data was skewed and in need of transformation. The GDP_per_Person data appeared to be heavily right-skewed, which is common for economic data where a few countries may have extremely high GDPs per person.

- **Applying a Logarithmic Transformation:** To address the skewness, I applied a logarithmic transformation to the GDP_per_Person variable. The log transformation is a common technique to compress large values and spread out smaller values, which helps in normalizing right-skewed data. By transforming the data in this way, I reduced the influence of extreme values, allowing for a more balanced analysis.
- **Checking the Transformed Data:** After applying the transformation, I checked the distribution of the newly created log_GDP_per_Person variable. The histogram showed that the data was now more normally distributed, making it better suited for analysis.

Results: Before the transformation, the GDP_per_Person variable was right-skewed, with a few countries having very high values. After applying a logarithmic transformation, the data became more normally distributed, reducing skewness.

Conclusion: The log transformation normalized the GDP_per_Person data, minimizing extreme values and making the data better suited for accurate statistical analysis.

Presentation

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Family name _____	Given name _____	Student no. _____
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